ELECTRICAL SYSTEM DESIGN AND ESTIMATION (RE)
Model Question Paper- Version A

Time : 3 Hour *Max.Marks : 75*

PART A

I. ***Answer any 5 questions. Each question carries 3 marks*** ***(5 x 3 = 15 Marks)***

1	Explain the scope of IS code IS 732.	M1.01	U
2	Explain the difference between MCB and ELCB.	M1.04	U
3	A shop 16m x 10m is illuminated with 200w incandescent lamps. If a CU of 0.8 and an MF of 0.75 are selected, and an illumination of 260 lux is required at the workplace, calculate the number of luminaires required.	M2.03	A
4	List the requirements of efficient street lighting.	M2.04	R
5	Explain the purpose of earthing in an electrical installation.	M3.01	U
6	List different basic components of grid-connected PV systems.	M4.03	R
7	Explain the need of a solar PV system for domestic application.	M4.01	U

PART B

Answer ONE question from each set. Each question carries 15 marks

II.

a)	A building has a total light load of 3000W consisting of 40 points and power load of 5000W consisting of 4 points. Determine the no.of sub circuits and no.of points needed in the installation. -8 Marks	M1.02	A
b)	Explain general rules and regulations for domestic wiring. -7 Marks	M1.01	U

OR

III.

a)	Explain any two protective devices used in domestic applications. -5 Marks	M1.03	U

b)	In a residential building, having 45 nos of light points , 10 fan points, 20 nos of 5 ampere plug socket, 6 nos of 15 ampere power plug socket and 1.5 HP single phase motor pump set (assume DOL starting). Calculate the total connected load, the no. of sub-circuits required, and select the conductors used for each sub-circuits. -10 Marks	M 1.02	A

IV.

a)	Explain the laws of illumination with neat diagrams. -5 Marks	M2.01	U
b)	A lamp giving out 1200 lm in all directions is suspended 8 m above the working plane. Calculate the illumination at a point on the working plane 6 m away from the foot of the lamp. -5 Marks	M2.03	A
c)	Explain the significance of maintenance factor and utilization factor in lighting. -5 Marks	M2.01	U

OR

V.

a)	A certain incandescent lamp hangs from the ceiling of a room. The illuminance received on a small horizontal screen lying on a bench 2m vertically below the lamp is 63.5 lux. Calculate illuminance at a point when the screen is moved horizontally a distance of l.5m along the bench -5 Marks	M2.03	A
b)	Explain different types of luminaires used in lighting. -5 Marks	M2.02	U
c)	Which are the energy conservation techniques in lighting? -5 Marks		U

VI.

a)	Calculate the load current and cable size of 20 HP motor of 415V, 50Hz, supply with 80% efficiency. -6Marks	M3.03	A

b)	In a Workshop (30mX15m) one 20hp, 400V, 3-ɸ Squirrel cage Induction Motor is to be installed. Prepare the estimate of the quantity of materials required . (Assume suitable values wherever necessary). -9Marks	M3.03	A

OR

VII.

a)	Calculate the rating of the starter of a 40HP Induction motor of 0.8pf, 75% efficiency. -5 Marks	M3.03	A
b)	Summarize pipe and plate earthing -5 Marks	M3.02	U
c)	Draw the single line diagram of a motor wiring -5 Marks	M3.03	U

VIII.

a)	Explain the general guidelines for installation of solar energy systems. -8 Marks	M4.01	U
b)	Draw a neat layout of a 2 KWp single phase grid connected solar PV system. -7 Marks	M4.03	A

OR

IX.

a)	Explain the design considerations of solar PV systems for domestic applications. -10 Marks	M4.04	A
b)	Explain any two different types of solar batteries -5 Marks	M4.01	U

Question wise Analysis – No of Question/Mark for each Question

Module	Hr/ module	(hi / ∑Hi) * 141	Type of questions					
			Part A (3 mark)		Part B (15mark)		Total	
			No	Marks	No	Marks	No	Marks
I	16	38.8	2	6	2	30	**4**	**36**
II	14	34.03	2	6	2	30	**4**	**36**
III	14	34.03	1	3	2	30	**3**	**33**
IV	14	34.03	2	6	2	30	**4**	**36**
Total							**15**	**141**

Question wise Cognitive level analysis– Mark for each Question

	Taxonomy Level		
	R	**U**	**A**
	Marks	**Marks**	**Marks**
Part A (3 Marks)	6	12	3

Part B (15 Marks)	0	55	65
Total (marks)=141	6	67	68

Question Wise Analysis

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.01	U	3	8
I.2	M1.04	U	3	8
I.3	M2.03	A	3	8
I.4	M2.04	R	3	8
I.5	M3.01	U	3	8
I.6	M4.03	R	3	8
I.7	M4.01	U	3	8
II. a	M1.02	A	8	20
II. b	M1.01	U	7	12
III. a	M1.03	U	5	12
III. b	M1.02	A	10	25
IV. a	M2.01	U	5	10
IV. b	M2.03	A	5	12
IV. c	M2.01	U	5	12
V. a	M2.03	A	5	12
V. b	M2.02	U	5	10
V. c	M2.02	U	5	10
VI. a	M3.03	A	6	12
VI. b	M3.03	A	9	25
VII. a	M3.03	A	5	12
VII. b	M3.02	U	5	15

VII. c	M3.03	U	5	15
VIII. a	M4.01	U	8	15
VIII. b	M4.03	A	7	18
IX. a	M4.04	A	10	20
IX. b	M4.01	U	5	15
Total			141	338

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ELECTRICAL SYSTEM DESIGN AND ESTIMATION (RE)
Model Question Paper- Version B

Time : 3 Hour *Max.Marks : 75*

PART A

I. ***Answer any 5 questions. Each question carries 3 marks*** ***(5 x 3 = 15 Marks)***

1	Summarize the concept of sub circuits	M1.02	U
2	Explain the reasons for providing the fuse on the line wire and never on the neutral wire.	M1.03	U
3	State and explain laws of illumination.	M2.01	R
4	List the main classification of building lighting arrangements.	M2.02	R
5	Summarize the requirement of earthing and list different types of earthing	M3.01	U
6	Explain the selection of cable and fuse ratings for motor installation.	M3.03	A
7	Draw a neat layout of a standalone solar power system.	M4.04	U

PART B

Answer ONE question from each set. Each question carries 15 marks

II.

a)	Explain the safety aspects that have to be considered while doing electrical dwelling in LV and MV installations. -6Marks	M1.02	U
b)	A three occupant building has to be electrified independently from a common energy meter. Design the distribution boards with accessories for each resident having 10 nos of light circuits, 6 nos of power circuits. -9 Marks	M1.03	A

OR

III.

a)	Explain with relevant examples, the selection of fuses and MCBs for different levels of circuit. -6Marks	M1.02	U

b)	Summarize are the general rules to be followed for internal wiring. -9 Marks	M 1.02	U

IV.

a)	An office 30mX15m is illuminated by 40W fluorescent lamps of lumen output 2700 lumens. The average illumination required at the workplace is 200lux.Calculate the number of lamps required to be fitted in the office. Assume coefficient of utilization to be 0.6 and depreciation factor 1.25. -7 Marks	M2.03	A
b)	With the help of neat wiring diagram, explain the working of mercury vapour lamp. - 8 Marks	M2.02	U

OR

V.

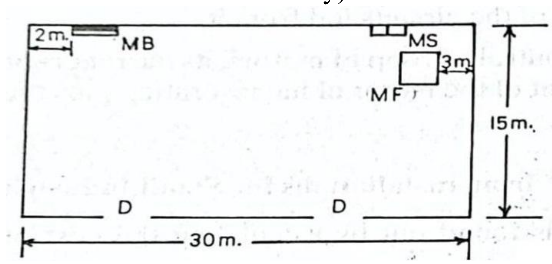
a)	With the help of neat wiring diagram, explain the working of sodium lamp -9Marks	M2.02	U
b)	Explain design considerations of a good lighting scheme. -6Marks	M2.03	U

VI.

a)	Draw the single line diagram of motor wiring. -5 Marks	M3.03	U
b)	With the help of neat layout explain standard pipe earthing -10 Marks		U

OR

VII.

a)	In a Workshop one 15hp(metric) 400V, 3-ф Squirrel cage Induction Motor is to be installed. Prepare the estimate of the quantity of materials required . (Assume suitable values wherever necessary).  -10 Marks	M3.03	A

b)	Prepare an estimation for the materials used in standard plate earthing. -5 Marks	M3.02	U

VIII.

a)	A residential building is having following connected loads. (i) An LED lamp of 40 W for 12 hours per day. (ii) A refrigerator of 80 W for 8 hours per day. (iii) A fan of 60 W for 6 hours per day. A 12V, 120W solar panel system is to be installed in the building. Calculate the number of solar panels required, rating and sizing of charge controller, inverter and batteries. -10 Marks	M4.04	A
b)	List important factors considered for installing a solar panel. -5 Marks	M4.02	U

OR

IX.

a)	Explain the design considerations of solar PV systems for domestic applications. -7 Marks	M4.04	U
b)	Draw a neat layout of 5 KWp three phase grid connected solar PV system -8 Marks	M4.03	A

Question wise Analysis – No of Question/Mark for each Question

Module	Hr/ module	(hi / ∑Hi) * 141	Type of questions					
			Part A (3 mark)		Part B (15mark)		Total	
			No	Marks	No	Marks	No	Marks
I	16	38.8	2	6	2	30	**4**	**36**
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IV	14	34.03	1	3	2	30	**4**	**33**
Total							**15**	**141**

Question wise Cognitive level analysis– Mark for each Question

	Taxonomy Level		
	R	**U**	**A**
Part A (3 Mark)	6	12	3

Part B (15 Marks)	0	76	44
Total (marks)=141	6	88	47

Question Wise Analysis

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.02	U	3	8
I.2	M1.03	U	3	8
I.3	M2.01	R	3	8
I.4	M2.02	R	3	8
I.5	M3.01	U	3	8
I.6	M3.03	A	3	8
I.7	M4.04	U	3	8
II. a	M1.02	U	6	10
II. b	M1.03	A	9	18
III. a	M1.02	U	6	10
III. b	M 1.02	U	9	18
IV. a	M2.03	A	7	18
IV. b	M2.02	U	8	15
V. a	M2.02	U	9	16
V. b	M2.03	U	6	16
VI. a	M3.03	U	5	18
VI. b	M3.03	U	10	20
VII. a	M3.03	A	10	20
VII. b	M3.02	U	5	20

VIII. a	M4.04	A	10	25
VIII. b	M4.02	U	5	18
IX. a	M4.04	U	7	25
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Total			141	338

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